2 Smart contracts

This chapter covers

- Understanding smart contracts
- Applying design principles to develop smart contracts
- Coding smart contracts with the Solidity language
- Running and transacting with smart contracts by using the Remix IDE
- Designing, developing, deploying, and testing smart contracts for two use cases

The smart contract is a significant component of the blockchain technology that has been instrumental in transforming a cryptocurrency framework into a trust framework enabling broad range of decentralized applications. This chapter provides details on the concept, design, and development of a smart contract, and also examines the power of executable code on the blockchain.

Structurally, a *smart contract* is a standalone piece of code similar to a class in an object-oriented program. It is a deployable module of code with data and functions. Functions serve the specific purposes of verification, validation, and enabling recording of the messages sent. A contract in the real world involves rules, conditions, laws, regulations to be enforced, criteria, contingencies, and items